

JUAN MANUEL RESTREPO

Junior Software Developer · Information Systems Technologist · Systems Engineering (in progress)

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PROFESSIONAL SUMMARY

Information Systems Technologist and Systems Engineering student with over 4 years of experience combining software development, quality assurance (QA) and agile methodologies (SCRUM). I build full-stack applications with VueJS, React, Node.js and both SQL and NoSQL databases, and complement my technical profile as a computer science teacher in bilingual International Baccalaureate (IB) programmes, designing projects in object-oriented programming, data structures and algorithms.

Self-taught and committed to continuous improvement, with logical thinking, strong technical documentation skills and problem-solving ability, applied both in production environments and in the classroom.

TECHNICAL SKILLS

Languages & frameworks: JavaScript, TypeScript, Python, Java SE, VueJS, Vuetify, React, Node.js, Express.js

Data & infrastructure: SQL (PostgreSQL, MySQL), MongoDB, Firebase, Prisma, Docker, REST APIs, JWT

Methodologies: SCRUM, agile development, SDLC, STEM/STEAM

QA & testing: Functional testing, bug reporting, software validation, technical documentation

Tools & other: GitHub, Google Workspace, Object-Oriented Programming (OOP), agile project management, Arduino

Languages: Spanish (native), English (B2+)

PROJECTS

Materiales Vera — Dump truck fleet management system

github.com/juanrex04/materiales-vera

Vue 3 · Express · MySQL · JWT · PDF

- Full-stack system with Admin/Driver roles and JWT authentication.
- 42-point pre-operational inspections, vehicle document tracking and PDF report generation.

School Timetable System — Optimization monorepo

github.com/juanrex04/sistema-horario-escuela

TypeScript · React · Node.js · Python · PostgreSQL · Docker

- Automatic school timetable generation (CSP problem) with 4 sections and asymmetric time slots.
- React frontend, Node.js/Express backend with Prisma, and a Python microservice using Google OR-Tools, orchestrated with Docker.

Real-Time Chat

github.com/juanrex04/chatFirebase

JavaScript · Firebase · Bootstrap · SCSS

- Real-time chat application using Firebase as a backend-as-a-service, with a custom Sass-styled interface.

Medical Dashboard

github.com/juanrex04/medicalApp_vuetify

Vue.js · Vuetify

- Medical dashboard built with Vuetify on Vue.js to explore UI components and design patterns for healthcare interfaces.

PROFESSIONAL EXPERIENCE

Computer Science Teacher — Primary Years Programme (IB)

Jul 2025 – Present

Colegio Cañaverales International School

- Designed visual programming sequences (MakeCode and Arduino) and computational logic integrated into digital platforms.
- Managed technology tools and online learning in a bilingual environment.
- Created digital and interactive teaching resources with a technical focus on content design.

Achievement: Developed integrated projects aligned with the Programme of Inquiry (POI), increasing student engagement with digital environments and fostering open-minded, international digital citizenship.

Computer Science Teacher — Diploma Programme (IB)

Aug 2021 – Jun 2025

Colegio Británico — The British School Cali

- Supervised IB Internal Assessment: design, coding, testing and documentation of student projects.
- Applied SCRUM in school projects aligned with industry standards.
- Taught OOP, data structures, databases and algorithms to grades 9, 10 and 11 in a bilingual environment.
- Developed technical and curriculum materials in English and Spanish for IB Computer Science.

Achievement: Improved the department curriculum plan by implementing programming-based integrated projects, and developed personalized follow-up plans according to each student's needs and IB requirements.

QA Tester / Junior Developer

Mar 2020 – Feb 2021

BM Software

- Performed functional QA, bug reporting and validation across the full software development cycle.
- Produced technical documentation: manuals, test records and reports.
- Developed minor features in JavaScript and Java SE under senior supervision.
- Collaborated in multidisciplinary teams using agile methodologies.

Achievement: Gained hands-on experience in production environments, solid documentation practices and SCRUM skills that I now apply and teach.

EDUCATION

B.Sc. in Systems Engineering (in progress)

Sep 2025 – Present

Universidad INCCA de Colombia

Diploma in Strategic Management and Innovation of Agile Projects

Dec 2019 – Jan 2020

Fundación Centro Colombiano de Estudios Profesionales

Technologist in Information Systems

Feb 2017 – Nov 2019

Fundación Centro Colombiano de Estudios Profesionales

CERTIFICATIONS

- Keys to Creating Your IB Programme: Primary Years Programme — *International Baccalaureate*, Feb 2026
- Applying the SCRUM Framework to Software Development Projects — *SENA*, Oct 2024
- Advisory for the Use of ICT in Training — *SENA*, Oct 2024
- Building Learning Activities Integrating ICT — *SENA*, Oct 2024
- Computer Science Cat. 1 (IB) — *International Baccalaureate*, Oct–Nov 2023
- Teaching Strategies in STEM Education — *ÚNICA*, Aug–Sep 2022
- Teaching Disciplines Through Computing — *Universidad ICESI*, Oct 2021
- Social-Emotional Learning in Action — *Universidad ICESI*, Oct 2021
- ICT Tools for Teaching Resources — *SENA*, Jul 2021
- Humane Pedagogy — *SENA*, Jul 2021